Mike Vincent
Los Angeles, CA | U.S. Citizen
https://mikevincent.dev

EDUCATION

University of Texas at Tyler
Master of Science, Industrial Engineering, 2015

The Pennsylvania State University
Bachelor of Science, Linguistics, 2011

EXPERIENCE

Lead Software Engineer, Search and Platform Infrastructure | PricewaterhouseCoopers (PwC)
2024 to Present | Los Angeles, CA

Founding engineer on payments, tokenization, fraud models, AI, and search infrastructure for Capital One and Discover.

- **Discover Global Network fraud machine learning models —** machine learning fraud and risk models: prohibited-transaction detection, feature pipelines, real-time inference, and serving.
- **Capital One Databolt —** vaultless, field-level tokenization in Go and TypeScript; AI models, analytics, and payments run on tokens and never see the real card number.
- **Capital One Eno —** NLU, entity, and intent models behind the account chatbot, moved from Kubeflow onto Capital One's enterprise AI platform.
- **CapitalOne.com Search —** hybrid search with autocomplete and indexing, surfacing results as a customer types.
- **Discover.com Search —** hybrid search over vector embeddings and BM25, scored on offline evaluation sets that measure answer quality.

Software Engineering Manager, Infrastructure | Cubic Transportation Systems
2021 to 2024 | San Diego, CA

Software engineering manager for transit-payment SaaS across five agencies — WMATA, NYC MTA, BART, LA Metro, and MBTA.

- **NYC MTA OMNY** — contactless iPhone fare payments with Apple; Kubernetes manifests and Docker containers for the payments app.
- **LA Metro TAP** — fare-payment backend as Docker and Kubernetes microservices with vertical, horizontal, and KEDA autoscaling on AWS and Azure.
- **Platform security** — mTLS with Istio and a Smallstep private certificate authority; CrowdStrike, Nessus, and Trend Micro across the fleet.
- **Release engineering** — Terraform, AWS, and GitHub Actions CI/CD, under PCI and SOC 1/2 compliance; production code in Python, Java, and JavaScript across AWS, Azure, and vSphere.
- **Ran a team of 18** across the US, India, and Germany supporting transit SaaS for millions of riders; on-call on PagerDuty, monitoring on Prometheus and Grafana.

Lead Senior Software Engineer | Cubic Transportation Systems
2019 to 2021 | San Diego, CA

Lead engineer building rider-facing chatbots, prediction, and the cloud platform behind transit fare services.

- **Chicago Metra** — rider chatbot built with Twilio and Azure Cognitive Services on AWS Lambda; riders check transit-card balances by phone, text, or email.

- **Urban Insights** — transit-demand machine learning and search in Jupyter and SageMaker, monitored with Prometheus and Grafana.
- **Build and release** — automated CI/CD for 20 Java apps with Jenkins, Ansible, Docker, Helm, Vault, Terraform, and Kubernetes on Azure; database migrations via Liquibase and SQL*Plus.

Senior Software Engineer | Cubic Transportation Systems
2017 to 2019 | Washington, DC

First DevOps engineer on Cubic's fare platforms, moving embedded and legacy systems to containers and cloud.

- **NextBus** — real-time bus-arrival prediction on AWS ECS and Lambda with TensorFlow, PySpark, CloudFormation, and Node.js.
- **First DevOps engineer** — led the transformation for a publicly traded company with Terraform, Kubernetes, and Docker; embedded fare CI/CD with Jenkins, Conan, and Wind River Linux.
- **Security and compliance** — PCI, SOC 2, ISO, and SOX audits; AWS and Azure hardened with Trend DSA, Nessus, and Cisco ASAv, monitored with Prometheus and Grafana.

Senior Software Engineer | EPAM Systems
2016 to 2017 | Washington, DC

Senior backend engineer at a digital agency, shipping search, payments, and mobile apps for enterprise clients.

- **Fannie Mae Collateral Underwriter** — Angular single-page app on Tomcat with JSON and REST APIs in Python; appraisers value a home on a mortgage.
- **Lockheed Martin Intranet Search** — enterprise search on headless Drupal; microservices on Docker and Kubernetes with Node.js and Python.
- **Release engineering** — Jenkins, TeamCity, and Sonatype Nexus CI/CD on AWS with Chef; PostgreSQL migrations with Liquibase and a zero-downtime multi-tenant switch.

Software Engineer | EPAM Systems
2015 to 2016 | Washington, DC

Backend engineer on a full-stack team building mobile apps and e-commerce for national brands.

- **Poison Control** — National Capital Poison Center iPhone and Android apps built with Cordova CLI, shipped via App Store, TestFlight, and Google Play.
- **Canadian Tire** Hybris store and the **U.S. Conference of Catholic Bishops** app on AWS, with RDS and ElastiCache behind Kubernetes and Docker.
- **Infrastructure** — Docker and Ansible on VMware ESXi with Ubuntu and CentOS behind Cisco ASA firewalls; provisioned with Vagrant and Chef.
- **Test automation** — Node, Karma, Jasmine, PhantomJS, WebDriver, and Protractor; JMeter and BlazeMeter performance tests; SVN-to-Git migration.

PROJECTS

Io Pictures Corporation - Founding Engineer
Architecture for rights, metadata, payments, storage, and digital distribution systems from zero to production.

The Archive of American Radio, Inc. - Founding Engineer
Architecture for search, RAG retrieval, metadata extraction, AI agent workflows, content processing, and API systems from zero to production.

SKILLS

Core: Software architecture, platform architecture, distributed systems, system design, site reliability engineering, SRE, incident response, on-call, backend services, APIs

Cloud: AWS, Azure, GCP, Kubernetes, Docker, Terraform, Ansible, CloudFormation, Packer, Helm, Istio, KEDA, Smallstep, GitHub Actions, Jenkins, TeamCity, Nexus, Vault, ECS, Lambda, RDS, ElastiCache, Linux

Languages: Go, Python, Java, TypeScript, JavaScript, SQL, PostgreSQL, Liquibase, Redis, Kafka

AI/ML: LLM, AI agents, RAG, machine learning, MLOps, vector search, hybrid search, BM25, ANN/KNN, entity extraction, intent classification, Kubeflow, TensorFlow, PySpark, SageMaker, Azure Cognitive Services

Security: mTLS, CrowdStrike, Nessus, Trend Micro, Cisco ASA, PCI, SOC 2, ISO, SOX

Search: Search indexing, ETL, offline evaluation, experiment design, Lucene, Solr, OpenTelemetry